

Multiple Choice (10 marks)

1. Message authentication is a service beyond
 - (a) Message Confidentiality
 - (b) Message Integrity
 - (c) Message Splashing
 - (d) Message Sending
2. How does a buffer overflow on the stack facilitate running attacker-injected code?
 - (a) By overwriting the return address to point to the location of that code
 - (b) By writing directly to the instruction pointer register the address of the code
 - (c) By writing directly to %eax the address of the code
 - (d) By changing the name of the running executable, stored on the stack
3. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet
4. Why would a ping sweep be used?
 - (a) To identify live systems
 - (b) To locate live systems
 - (c) To identify open ports
 - (d) To locate firewalls
5. Which of the following is not a transport layer vulnerability?
 - (a) Mishandling of undefined, poorly defined variables
 - (b) The Vulnerability that allows "fingerprinting" & other enumeration of host information
 - (c) Overloading of transport-layer mechanisms
 - (d) Unauthorized network access

True / False (10 marks)

Answer True or False

6. True or False: Message integrity ensures that data arrives at the receiver exactly as it was sent, without any alteration during transmission.
7. True or False: Wireless security involves measures to protect computers and data from unauthorized access via wireless networks.
8. True or False: Nmap categorizes port states as active, inactive, and standby based on network conditions.
9. True or False: Message integrity guarantees that the content of a message remains confidential from all parties involved.
10. True or False: The stack is primarily used for storing local variables during function execution.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the perimeter of a rectangle.
12. Write a function to insert an element before each element of a list.

End of Paper